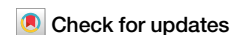


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Country's wealth is not associated with domestic control of metal ore extraction



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Countries increasingly rely on metal ores for the energy transition but face potential supply constraints. Here we combine multi-region input–output analysis with corporate ownership data from 2000 to 2022 for ten widely used metals, and visualize the results using Sankey diagrams for 159 countries plus four world regions. We find no association between a country's wealth and the share of its locally extracted ores under domestic control, highlighting the highly globalized nature of the mining sector. Similarly, overseas mine ownership does not translate into direct material flows back to the controlling region, indicating that market forces can overshadow corporate influence. Despite no measurable trade shifts, foreign ownership remains a potential lever for supply chain disruptions. By integrating corporate control into criticality assessments, policymakers can identify future vulnerabilities and strengthen resource security. Enhanced data transparency is also essential for building resilient supply chains.

Countries increasingly rely on metal ores to meet global energy transition targets, but face potential supply constraints. Here we combine multi-region input–output analysis with corporate ownership data from 2000 to 2022 for ten widely used metals. We find no association between a country's wealth and the share of its locally extracted ores under domestic control, illustrating the highly globalized nature of the mining sector. Similarly, overseas mine ownership does not translate into direct material flows back to the controlling region, suggesting that market forces can overshadow corporate influence. Despite the absence of measurable trade shifts, foreign ownership remains a potential lever for supply chain disruptions. By integrating corporate control into criticality assessments, policymakers can more effectively identify future vulnerabilities in raw material supply. Improved data transparency is essential to better navigate complex resource dependencies and build resilient supply chains.

Raw materials are fundamental to society¹ and critical to the global energy transition^{2–5}. As most nations rely on raw material imports to meet their needs, they face important dependencies on foreign sources. Criticality assessments are therefore used to guide decisions on material selection, product design, investments, or trade agreements⁶. Two dimensions are central to these assessments (i) the vulnerability of a region or consumer to supply disruptions, and (ii) the likelihood for the occurrence of such disruptions⁷. Regional concentration of imports is often used as a proxy for the likelihood of disruption⁶. However, as mining is controlled by different companies that can behave differently within the same country⁸. It is thus necessary to complement existing criticality analyses with an examination of “the locus of corporate

control”, i.e. where the headquarters of the companies controlling production are based⁹.

This locus of corporate control has been studied for cobalt over the period 1975–2018⁸, for 6 metals over the period 1985–2018⁹ and for Chinese control in Africa from 1995 to 2018¹⁰. These studies provide valuable insights into corporate influence over resource extraction but they have not analysed the impact of control on trade. Sun and colleagues compared overseas investments with domestic demand for lithium, nickel, cobalt, and platinum in 2019, revealing an overlap between investment partners and trade flows¹¹. However, they did not carry out any statistical analysis, so it is unclear whether the overlap is a random effect or if there is a real trend of foreign-owned companies exporting metals to their countries of origin. Moreover, their study focused exclusively on ores, whereas foreign-owned companies might refine metals locally before exporting them.

Global value chains often involve multiple countries, each specializing in distinct production stages. In particular, many transformation steps occur between metal mining and final consumption¹². To efficiently assess vulnerability, the first dimension of criticality assessments, it is thus crucial to account for the metals embodied in final consumption, not just in direct imports as done previously in the literature^{13,14}. This can be accomplished through environmentally-extended input-output analysis (EE-IOA)¹⁵. EE-IOA describes interdependencies between different sectors within an economy^{16–18}. It tracks environmental pressures along global supply chains and allocates these pressures to the final consumption activities that ultimately drive production¹⁹. EE-IOA allows the calculation of “consumption-

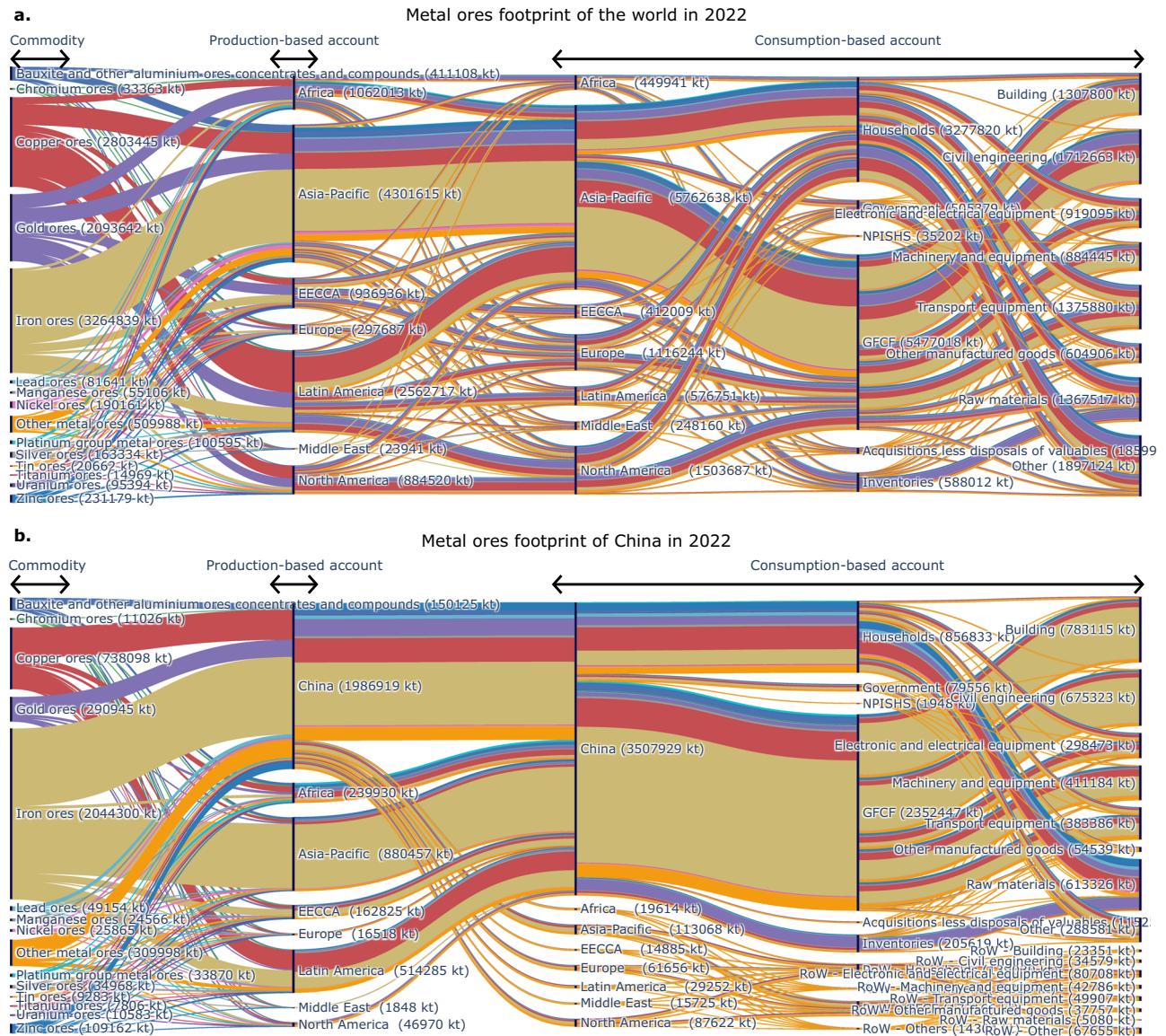


Fig. 1 | Metal ores footprint in 2022. The first step corresponds to the commodity, the second step to the production-based accounts and the last three steps to the consumption-based account by region, final consumer, and final consumption

sector. **a** Global picture. **b** China focus. Any flow represented in **b** is either produced in China, or embodied in China's final consumption. The colors correspond to the first step of the diagram.

based" environmental accounting, which allocates environmental impacts to the place where goods are consumed, rather than where they are produced ("production-based" account). EE-IOA has been used to calculate the metal footprints of different regions^{20–22}, but without taking into account the locus of mining control. A recent study combined EE-IOA with multinational enterprise data to trace material flows along global value chains, but all metals were aggregated, and the nationality of foreign-owned firms was not examined²³.

To inform future criticality studies, we aim to analyse regional dependence on imported metal ores, both directly and indirectly (via metals embodied in final consumption), distinguishing between the countries which produce the metals and those which own the mines. We utilize the multi-regional input-output database GLORIA^{21,24} to trace the value chains and S&P Capital IQ Pro²⁵ for the specific mine control data. Although GLORIA covers 14 commodities over period 1990–2022, the main analyses of this study are limited to the ten metals—copper, nickel, lead, zinc, tin, iron, aluminium, uranium, gold, and silver—that overlap with S&P control data, over the period 2000–2022. For the 159 countries plus 4 world regions in GLORIA, we visualize the results using Sankey diagrams. In these

diagrams, flow are proportional to their values, allowing us to trace material movements and ownership.

We address four key questions: Does the wealth of a country/region influence the share of ownership (control) over their domestic extraction? Do import dependencies—both from a producing-country and controlling-country perspective—vary when considering direct imports or indirect imports (i.e. embodied in final consumption)? Do the countries that control foreign production import the metals they extract abroad? What are the regions with the highest potential for disrupting metal supplies via their mine ownership?

Results and Discussion

Mapping the value chain

Sankey diagrams are effective tools for visualizing complex supply chains and identifying the largest flows^{26–28}. Figure 1a compares global metal production-based account with global metal consumption-based account. The first step of the diagram shows the commodity type, while the second step represents the geographic disaggregation of the production-based account (i.e. in what region is the metal extracted). The transition from steps

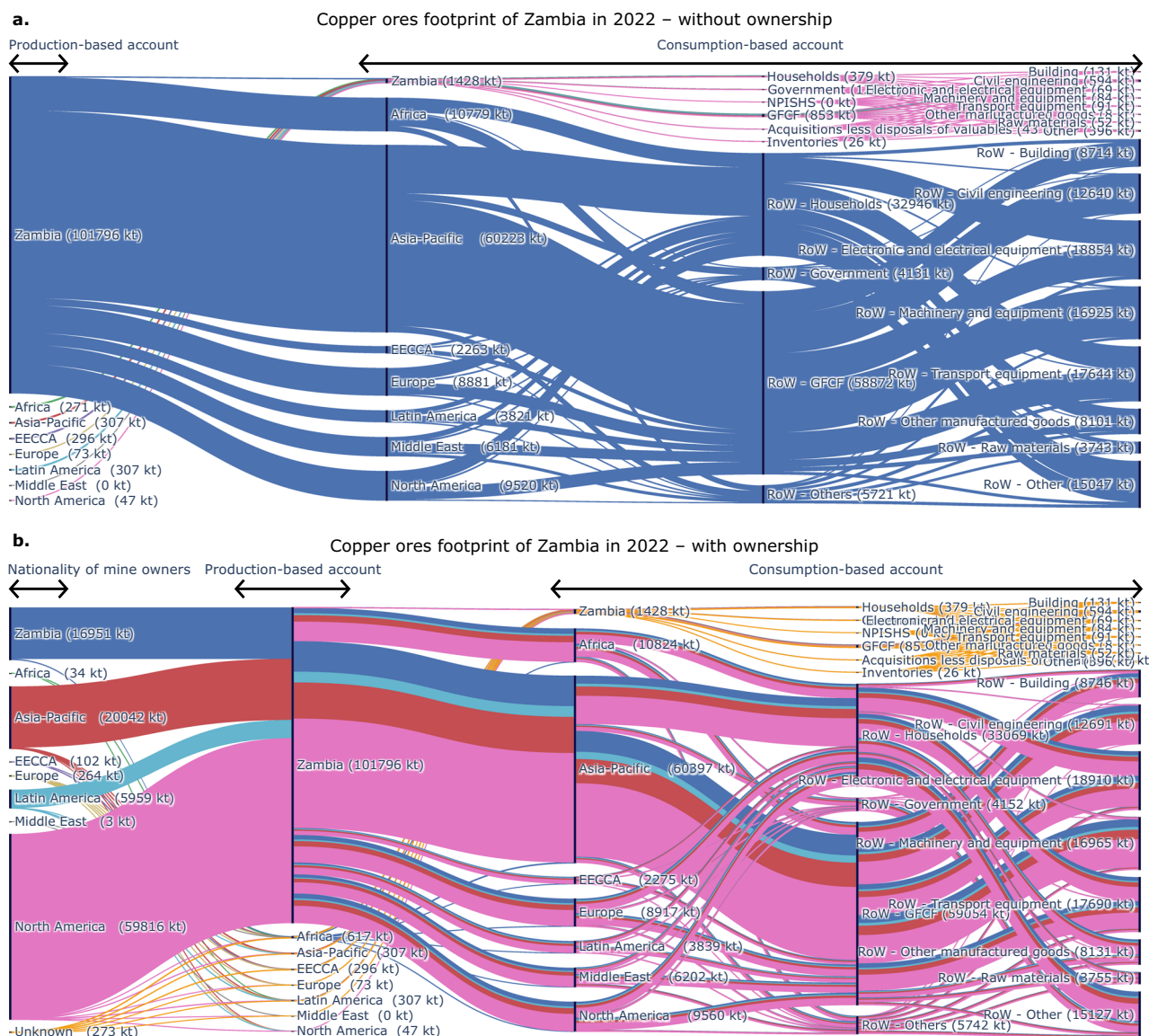


Fig. 2 | Sankey diagrams of Zambia's copper footprint in 2022. a The first step represents the production-based account (extraction). The three later steps represent the copper embodied in final consumption represented either by region, type of consumer, or final demand sector. Any flow represented in **a** is either produced in Zambia, or embodied in Zambia's final consumption. **b** A step is added on the left

hand side of the diagram to represent the nationality of the companies that control Zambia's mines. Any flow represented in **b** is either controlled by Zambian-controlled companies, produced in Zambia, or embodied in Zambia's final consumption. **a, b** The colors correspond to the first step of the diagram.

2 to 3 uses Leontief's inverse (see Methods) to go from production-based to consumption-based accounting, revealing, for example, that North America consumes 70% more metal ores than it produces. The right-hand side of the diagram represents the consumption-based account, disaggregated by region of consumption (step 3), by consumer type (step 4) or by final consumption sector (step 5). The final consumption sector is disaggregated between 8 types of goods or services that are bought by final consumers or for capital creation. For example, at the global level, 1.4 Gt of metal ores have been used in 2022 (along the whole value chain) to produce transport equipment. It is important to note that the "raw materials" node represents the metal ores embodied in the final consumption of products that are directly purchased by end users. Although consumers do not buy raw metal ores, they do purchase items such as cement, glass, and clay building materials, which incorporate these metal ores in their production processes.

We also built diagrams that zoom into specific countries or regions. For instance, Fig. 1b maps the metal ores flows related to China in 2022. Each flow represents either ores produced in China or embodied in China's final consumption. The diagram reveals that China's consumption-based

account is 75% higher than its production-based account. For the 15 commodities in GLORIA, the period 2000–2022, and the 163 GLORIA regions, these Sankey diagrams are available on the web application associated with this paper: sankey-dashboard.efficiency.org.

Figure 2a represents Zambia's copper footprint in 2022. This is a striking example of a developing country whose footprint is very small (1.4 Mt, e.g. 0.07 t per capita) but that extracts a massive amount of ores (101 Mt, e.g. 5.1 t per capita). Even though it produces a lot, the copper ores embodied in Zambia's final consumption are imported from other regions. This is due to the high level of complexity of global value chains²⁹ and to the fact that Zambia in particular has not yet developed its downstream processing capacity³⁰. To have a broader understanding of Zambia's copper industry, data on the control of the mines is required. Here, we use control in the sense of Ericsson, i.e. "the possibility to act decisively on strategically important issues rather than day-to-day influence over a company"²⁹. We use data from the S&P Capital IQ Pro database, which uses ownership data, collected directly from mining companies, as a proxy for control²⁵. If a mine produces 10 kt of copper ores in Zambia and is owned at 40% by a company whose

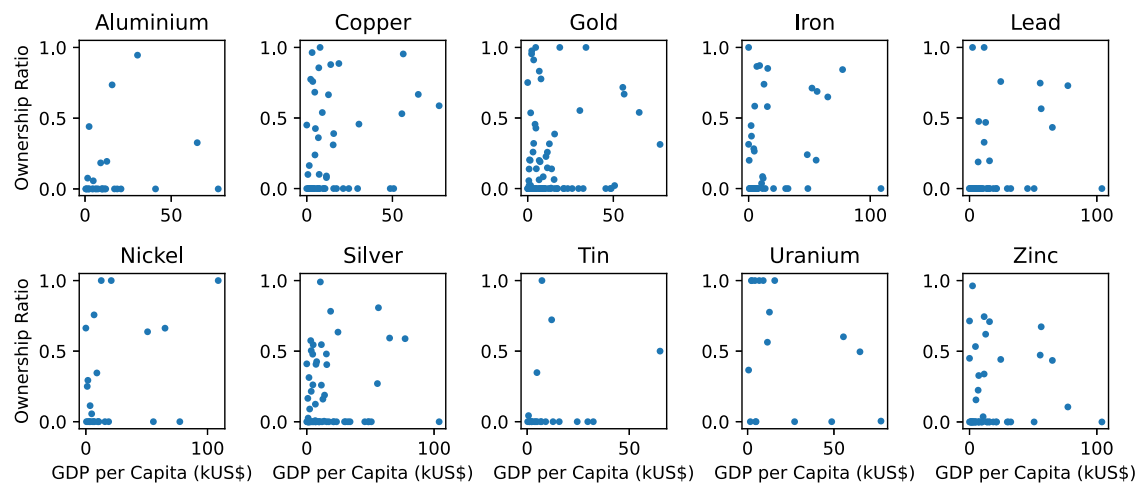


Fig. 3 | Ownership ratio as a function of GDP per capita in 2022. The ownership ratio is the share of a country's mine production controlled domestically, divided by its total mine production.

headquarter is in Canada, then Canada is considered to control 4 kt of that mine's production. This is in line with previous work in the literature^{8–11}. Adding an ownership layer before the production-based step gives a very different point of view of Zambia's copper production. On Fig. 2b, the first step shows that only 17% of the copper extracted in Zambia is controlled by Zambian companies. The dominance of Zambia's mining sector by foreign-owned companies, which focus solely on extraction without expanding into downstream activities, has limited the economic benefits for the local population³⁰. Profits from mining largely flow to international investors via stock exchange listings, while the primary economic benefit to Zambia mainly comes from royalties and taxes³⁰. The DRC-Zambia battery plant project is an example of a promising initiative to add more value added locally³¹. The Sankey diagrams with the ownership layers for all 10 commodities and 163 regions are available in the second tab of the web application: sankey-dashboards.refficiency.org.

To determine if Zambia's limited control over its own mines is related to its development level, we calculated an ownership ratio—the share of a country's mine production controlled domestically, divided by its total mine production—and compared it to Gross Domestic Product (GDP) per capita. Figure 3 shows that for the ten commodities studied, no correlation was found between ownership ratio and GDP per capita. This can be explained by the globalized nature of the mining industry, where mining companies from a few industrialized countries control most of the world production and compete intensely, including in their own countries when mining is still taking place there⁹.

Import dependencies

Many regions have an ownership ratio of 0. This indicates that while these regions may appear independent when considering production alone, they are not when ownership is taken into account. To assess the extent to which regions rely on imported ores, we introduce four dependency measures that capture two distinct dimensions—production and ownership—and that are calculated in two ways (aggregated and “real”).

Aggregated dependency measures quantify the overall share of a region's final consumption that is not met by domestic sources. The production-based aggregated dependency compares total domestic ore production to the ores embodied in final consumption, while the ownership-based aggregated dependency compares the ore production controlled by domestic companies to the final consumption. In our analysis (Fig. 4a), a region is considered 100% dependent if it produces nothing (in the production-based measure) and has no domestic control (in the ownership-based measure), and 0% dependent if domestic production or control exceeds the ores embodied in final consumption. Figure 4a shows

that dependency values vary considerably depending on whether production or control is considered, underscoring the importance of examining both dimensions.

However, aggregated measures may overstate domestic self-sufficiency because they include domestic production that does not actually reach final consumption. To provide a more realistic picture, we calculate “real” dependencies. The real production dependency considers only the portion of domestic production that ends up in the region's final consumption, while the “real” ownership dependency considers only the fraction of final consumption that is met by production controlled by domestic companies. Figure 4b shows these real dependency measures, and the difference is striking: many regions that appear less dependent on imports when using aggregated measures are in fact highly reliant on imports when only effective domestic flows are considered.

For clarity, regions that are 100% dependent on all four metrics are omitted from Fig. 4. The number of such regions varies by commodity—for example, there are 129 regions (80%) for aluminium, 94 (58%) for copper, 57 (35%) for gold, 100 (62%) for iron, 107 (66%) for lead, 125 (77%) for nickel, 84 (52%) for silver, 136 (84%) for tin, 139 (86%) for uranium, and 97 (60%) for zinc. On average, across all commodities, 76% of regions exceed 90% dependency based on aggregated production and 85% based on aggregated ownership, while for real dependencies these shares increase to 87% and 94%, respectively. For detailed definitions and mathematical formulations of these indicators, please refer to the Methods section.

Impact of control on trade

Metals extracted by companies controlled by a given country do not find their way into the final consumption of that country. This does not necessarily mean that control has no influence on initial trade. For example, Japan invested in Australia's Lynas rare earth mine in exchange of priority supply rights³². Company owners could choose to lock-in production and prioritize exports to their own countries. To explore this, we tested whether control by country B over the mines of country A impacts trade from A to B. Both control and trade were measured in tons of ores. For each commodity, we found no correlation between control and direct ores exports from A to B (first column of Table 1). However, if a mine is controlled by region A, it might not export ores right away, but rather transform ores into metals locally and then export to B. On the second column of Table 1, we compared the exports of metals from A to B, selecting only the amount of metal that was produced using ores extracted in country A, with the control from B in A, and again found no correlation. Finally, we carried out this analysis using indirect exports of ores from A to B, using the consumption-based account of B and again found no correlation (third column of Table 1). While Sun

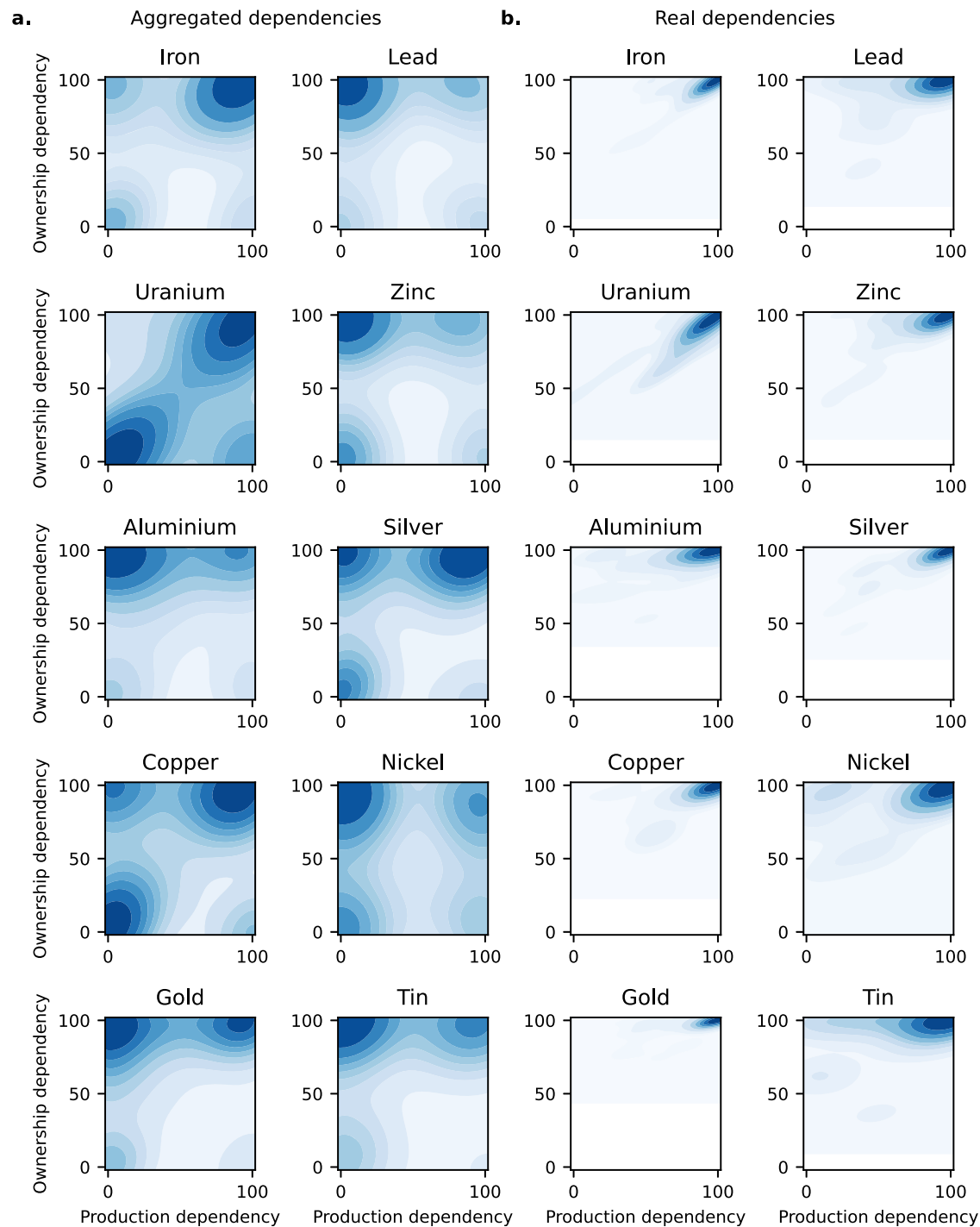


Fig. 4 | Import dependencies from an ownership perspective and from a production perspective in 2022. a Aggregated dependencies: $(1-\alpha)*100$, with α the total amount of ores controlled (respectively produced) by a country divided by the amount of ores embodied in its final consumption (consumption-based account). **b** “Real” dependencies: $(1-\beta)*100$, with β the share of a country’s consumption actually controlled (respectively produced) by that country, or the share of its consumption actually extracted by companies controlled by that country.

a, b Values < 0 are set to 0 and values > 100 are set to 100. Regions whose four dependencies are equal to 100 are not represented as this figure focuses on the difference between methods. Each subplot’s color map represents its local density distribution (calculated independently using `sns.kdeplot`), so the colors facilitate pattern comparison between “Aggregated” and “Real” dependencies rather than indicating absolute density values.

and colleagues suggested a potential relationship between overseas investments and trade flows, our statistical analysis indicates that their observed overlap may have been coincidental¹¹.

These results align with specific studies on Chinese investments. Although China’s investments in natural resources companies have aroused concerns of “locking up” supplies, they have actually contributed in the

diversification and increase in competitiveness of the global value chain^{33,34}. Overall, the highest correlation coefficient ($R^2 = 0.27$) was for indirect uranium flows. Given uranium’s critical importance, it is logical that countries reliant on it would invest in mining operations. Kazakhstan, for example, requires technology transfer from its partners in exchange for access to its uranium mines³⁵.

Table 1 | Squared correlation coefficients between trade and control in 2022

	Ore exports	Metal exports	Indirect ore exports
Iron ores	0.00	0.04	0.02
Uranium ores	0.00	NaN	0.27
Aluminium ores	0.01	0.00	0.06
Copper ores	0.04	0.02	0.24
Gold ores	0.01	0.03	0.06
Lead ores	0.02	0.03	0.09
Zinc ores	0.02	0.00	0.03
Silver ores	0.06	0.02	0.19
Nickel ores	0.00	0.05	0.00
Tin ores	0.04	0.00	0.08

Squared correlation coefficients (R^2) between control by country B over mines in country A and trade flows from A to B in 2022. Three trade measures are shown: direct ore exports, metal exports (metals produced from A's ores), and indirect ore exports (ores embodied in B's final consumption). No significant correlations are observed.

Our findings reveal that higher GDP per capita does not necessarily translate into stronger control over domestic resources, nor does holding notable foreign mining assets guarantee direct material flows back to the controlling nation. This underscores the complexity of global supply chains, where market dynamics, regulatory environments, and geopolitical considerations can overshadow simple economic metrics. Beyond corporate control, factors such as political ties, trust, linguistic proximity, and institutional linkages also play a key role in shaping trade flows^{36–38}. For policymakers and industry stakeholders, these insights illustrate that reliance on GDP or ownership-based strategies alone may be insufficient for ensuring secure supply. Instead, robust criticality assessments should integrate multiple dimensions—economic, geopolitical, and corporate governance factors—to gain a more complete understanding of vulnerabilities and resilience in raw material value chains.

Although control of foreign production –or of local production– does not have a major impact on trade today, it still grants potential power to countries. A striking example of a market disruption is the 2010–2012 rare earth crisis, during which China reduced its rare earth export quotas by 40%³⁹. Assuming countries can exert control over their companies operating abroad –which is not always certain– similar disruptions could arise from foreign domestic investments. To explore this potential more comprehensively, Fig. 5 shows the evolution of production-based accounts, consumption-based accounts, and controlled production from 2000 to 2022 for 11 countries that rank among the top five in at least one of these dimensions.

Resource-rich exporters such as Australia, Brazil, and Chile increased their domestic extraction by 301%, 216%, and 153%, respectively. In Australia and Brazil, final consumption also rose by 223% and 146%, while the production controlled by companies increased by 269% (from 404 Mt to 1088 Mt) and 263% (from 157 Mt to 413 Mt). In Chile, however, final consumption dropped to 48% (from 77 Mt to 37 Mt) of its 2000 value despite increases in both extraction (from 542 Mt to 828 Mt) and controlled production (from 183 Mt to 262 Mt), suggesting a decoupling between production and local use.

China exhibits the most pronounced changes. Domestic extraction grew by 343% (from 511 Mt to 1753 Mt) and final consumption by 504% (from 634 Mt to 3194 Mt), while the production controlled by companies surged by 1686% (from 76 Mt to 1275 Mt). Although this growth in controlled production is remarkable, its absolute volume in 2022 (1275 Mt) remains lower than both total domestic extraction (1753 Mt) and final consumption (3194 Mt). This outcome reflects a strategic emphasis on acquiring external mining assets to support booming domestic demand,

even though controlled production still does not represent the bulk of China's overall metal flows.

India and Russia display relatively balanced growth. India's domestic extraction increased by 288% (from 96 Mt to 275 Mt), final consumption by 332% (from 134 Mt to 445 Mt), and controlled production by 170% (from 66 Mt to 112 Mt). Russia's domestic extraction rose by 144% (from 231 Mt to 332 Mt), final consumption by 299% (from 66 Mt to 198 Mt), and controlled production by 155% (from 123 Mt to 189 Mt), indicating steady expansions across all dimensions.

The United Kingdom's final consumption declined by 27% (from 162 Mt to 117 Mt) while its controlled production increased by 239% (from 326 Mt to 778 Mt). Japan and South Korea both have almost no domestic extraction, but Japan registered increases in controlled production of 369% (from 47 Mt to 175 Mt) While South Korea's controlled production stayed very low (from 0 Mt to 1 Mt), alongside moderate rises in final consumption.

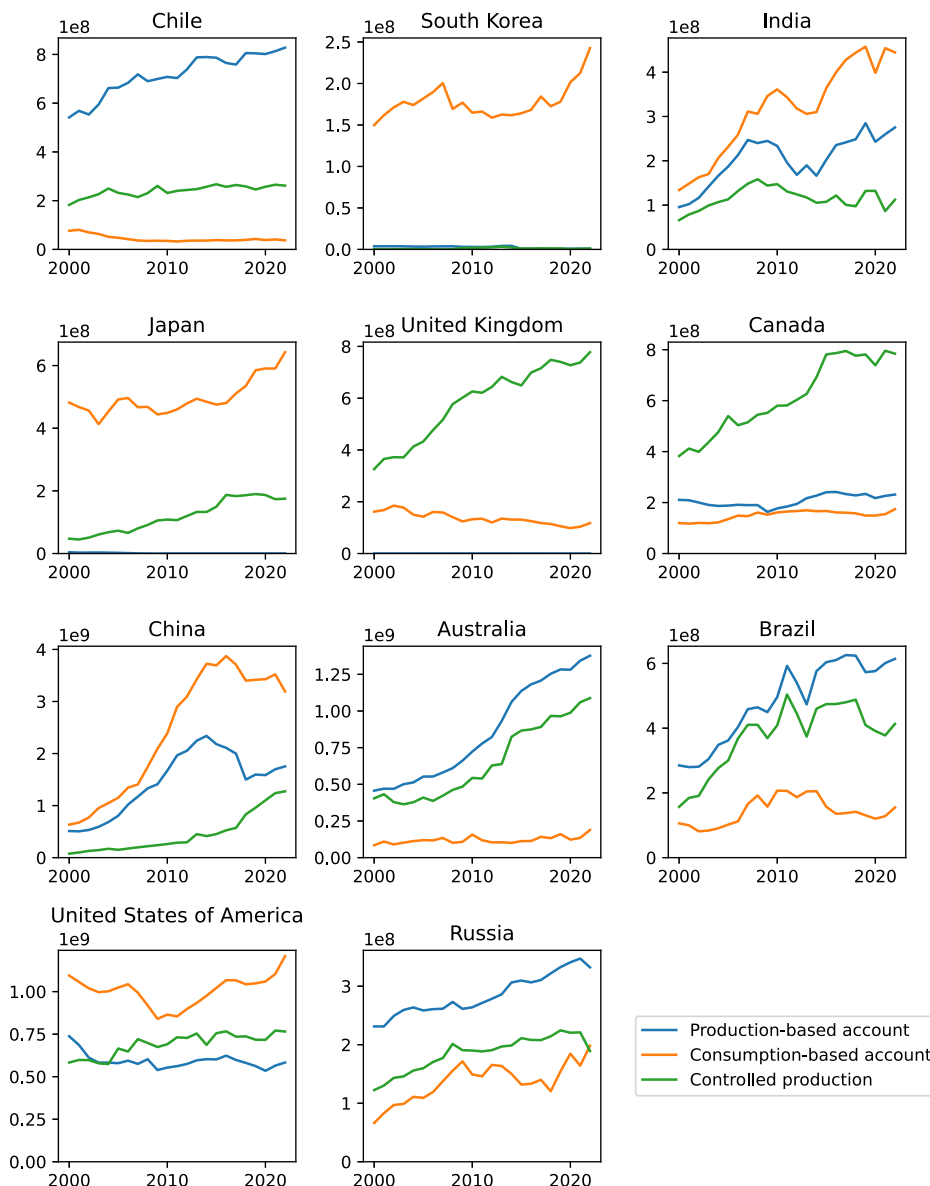
In contrast, the United States experienced a 21% decline in domestic extraction (from 738 Mt to 583 Mt), a 10% rise in final consumption (from 1094 Mt to 1209 Mt), and a 31% increase in controlled production (from 583 Mt to 766 Mt).

For year 2022, Table 2 lists the countries that control more than 10% of global production for various commodities (column 1) and compares this with their share of the global consumption-based account and production-based account (columns 2 and 3). The countries appearing in the table include China (five times), Canada (four times), the United Kingdom (three times), the United States (three times), Australia (once), Brazil (once), Kazakhstan (once), and Hong Kong (once). While the production controlled by Kazakhstan (for uranium), Australia (for iron), and Brazil (for iron) is much higher than their consumption-based account, it is lower than their production-based account. For the United Kingdom (aluminium, copper, iron), Canada (copper, gold, silver, uranium), and Hong Kong (uranium), the amounts controlled are considerably higher than both the consumption-based and production-based accounts. Historically industrialized nations (e.g., the United Kingdom and Canada) continue to exert a disproportionately high degree of ownership-based control relative to both their domestic extraction and consumption. Although the USA and China control substantial amounts, their share of control remains lower than the share of their consumption-based account. Although China's share of control has increased over time¹⁰, Chinese companies are far from dominating global mine production in 2022. The historical mining countries still control the majority of production.

The main limitation of the present analysis is the important share of production for which the ownership control is unknown: 50% for aluminium, 13% for copper, 32% for gold, 19% for iron, 43% for lead, 48% for nickel, 34% for silver, 77% for tin, and 27% for Zinc. Chinese ownership may have been underestimated as Chinese companies, including publicly traded ones, tend to be less transparent in their reporting compared to their multinational counterparts⁸. This is even more pronounced for companies predominantly controlled by Chinese interests¹⁰. Without standardized and comprehensive reporting, policymakers and industry stakeholders are left to navigate a landscape where the real influence of corporate control is masked by data gaps. This lack of transparency not only undermines the effectiveness of criticality assessments but also limits the ability of governments to regulate the broader social and environmental impacts of mining activities, both domestically and abroad. Consequently, there is an urgent need for enhanced regulatory frameworks that mandate clearer, more consistent disclosure of ownership structures, thereby enabling more accurate risk evaluations and fostering greater accountability across the global mining sector.

Another limitation is the theoretical nature of country control through ownership. Home countries often fail to regulate the impacts their mining companies have on host communities, which could also apply to their control over trade partners⁴⁰. Even for China, which is believed to exert strong control over its companies, the extent of state control varies depending on the type of company^{10,41}.

Fig. 5 | Time-series evolution of production-based, consumption-based, and ownership-based metal accounts (2000–2022) for selected countries. The units are metric tonnes. The same figure in tonnes per capita is available in the FigShare folder.



Methods

Input output analysis

GLORIA. Multi-Region Input-Output (MRIO) analysis is a powerful tool for tracing the flows of goods, services, and raw materials through complex global supply chains. By linking production and consumption activities across different regions, MRIO models provide a comprehensive view of how economic activities in one region can impact resource extraction, environmental degradation, and socioeconomic outcomes in others. These models are particularly valuable for assessing the environmental and economic footprints of consumption and production, offering insights into the resource dependencies and environmental impacts of global trade. MRIO analysis enables the quantification of direct and indirect material use, greenhouse gas emissions, and other environmental pressures associated with the entire lifecycle of products, from raw material extraction to final consumption. This level of detail is crucial for policymakers aiming to implement effective strategies for sustainable development, resource efficiency, and climate change mitigation. Here, we use the release 059 of the GLORIA global environmentally-extended MRIO database²¹, constructed in the Global MRIO Lab²⁴.

This database offers several advantages that make it particularly well-suited for this study. One of its key strengths lies in its ability to provide a comprehensive, continuous time series of global material footprints, covering detailed supply chains for a wide range of materials. Unlike previous MRIO databases, GLORIA addresses critical gaps by offering timely, detailed, and reliable data²¹. It utilizes an innovative computational collaborative research platform, integrating domestic and international trade data with material extraction information for each country²⁴. This comprehensive integration allows for precise tracing of materials through multiple stages of the supply chain, from ore extraction to final consumer products. GLORIA's detailed sector structure is specifically tailored to capture the intricacies of material supply chains, avoiding common aggregation issues that can lead to implausible attributions in other MRIO databases²¹. By applying engineering knowledge to material flows, GLORIA can more accurately reflect the actual distribution of materials across industries and countries²¹. Additionally, its participatory approach enables countries to contribute their own domestic extraction data, enhancing the database's accuracy and relevance. Furthermore, GLORIA's integration into UNEP's SCP-HAT tool ensures accessibility for policymakers and stakeholders, making it a vital resource for monitoring progress towards the

Table 2 | Countries controlling more than 10% of the production for the ten commodities studied

Commodity	Region	Control	Consumption	Production
Aluminium	United Kingdom	14%	1%	0%
	Unknown	50%	0%	0%
Copper	Canada	11%	2%	3%
	China	12%	24%	13%
	United Kingdom	11%	1%	0%
	United States of America	14%	17%	11%
	Unknown	13%	0%	0%
Gold	Canada	20%	2%	3%
	United States of America	12%	18%	10%
	Unknown	32%	0%	0%
Iron	Australia	20%	1%	30%
	Brazil	11%	2%	13%
	China	22%	57%	28%
	United Kingdom	10%	1%	0%
	Unknown	19%	0%	0%
Lead	China	28%	59%	56%
	Unknown	43%	0%	0%
Nickel	China	10%	14%	6%
	Unknown	48%	0%	0%
Silver	Canada	13%	2%	1%
	United States of America	13%	24%	8%
	Unknown	34%	0%	0%
Tin	Unknown	77%	0%	0%
Uranium	Canada	17%	2%	2%
	Hong Kong	11%	0%	0%
	Kazakhstan	35%	29%	62%
Zinc	China	28%	46%	41%
	Unknown	27%	0%	0%

When the share of unknown control over mines was higher than 10%, we also added 'Unknown' in the table. See file 'coverage' in the FigShare for more information on coverage rates by the S&P database.

Sustainable Development Goals related to resources, particularly goals 8.4 and 12.2²¹.

The development of the Sankey web application associated with this paper follows this same philosophy of making data comprehensive and accessible. By providing an intuitive visual interface, it allows users to interact with and understand complex material flows in a more accessible manner. The application democratizes data access, enabling stakeholders to explore detailed supply chain information and derive insights that can inform decision-making.

In the following, we use classic input-output formalism. The reader is referred to ref. 16 for an overview of input-output analysis, and to ref. 21 for a detailed presentation of GLORIA. The transaction matrix T has two row levels: a production region r in $\{1, \dots, 164\}$, a production sector i in $\{1, \dots, 120\}$; and two column levels: an intermediate consumption region s in $\{1, \dots, 164\}$, an intermediate consumption sector j in $\{1, \dots, 120\}$. T is thus the matrix of elements T_{ij}^{rs} with a dimension of 19680×19680 . The Leontief inverse L has by definition the same dimension and indexes as T . The final demand Y has

rows of the same dimension as the columns of L , and columns made of the final consumption region t in $\{1, \dots, 164\}$ and the final demand category k in $\{1, \dots, 6\}$. Y is thus the matrix of Y_{jk}^{st} with a dimension of 19680×984 . The factor of production coefficients for a given metal ore are given by q_i , a vector of elements q_i^r and of same dimension as the lines of L . Even though GLORIA comprises 164 regions, we refer to 163 regions in the main text as we excluded "DR Yemen (Aden)" which only had data for years 1990, 1992 and 1993. "Yemen Arab Republic/Yemen (1990/1991)" is one of these 163 regions.

In addition to the classic input-output formalism described above, it is important to emphasize that the GLORIA database overcomes the common limitation of low data granularity in MRIO analyses by providing a disaggregated view of metal ores. In contrast to many MRIO databases that group all metals together in a single category, GLORIA includes separate extensions for each metal ore. Specifically, the database offers extensions for Iron ores, Silver ores, Bauxite and other aluminium ores – gross ore, Gold ores, Chromium ores, Copper ores, Manganese ores, Other metal ores, Nickel ores, Lead ores, Platinum group metal ores, Tin ores, Titanium ores, Uranium ores, and Zinc ores, with production values expressed in tonnes. For a similar analysis for other metal ores, see our ongoing work in ref. 42.

Moreover, each of these extensions is associated with a distinct production sector. For example, GLORIA distinguishes between sectors for Iron ores (sector 28), Uranium ores (sector 29), Aluminium ore (sector 30), Copper ores (sector 31), Gold ores (sector 32), Nickel ores (sector 34), Tin ores (sector 35), and Other non-ferrous ores (sector 36). The only exception is the grouping of Lead, Zinc, and Silver ores into a single sector (sector 33). As a result, the production coefficients q_i^r in our analysis are specific to each metal ore extension, thereby enabling a material-specific tracing of footprints from extraction to final consumption. This disaggregation is a key strength of the GLORIA database and forms the basis for our material-specific analysis, addressing concerns about the typical aggregation found in MRIO data.

Ownership. To capture the influence of corporate control on metal flows, we integrate mine ownership data into GLORIA. For ownership data, previous research has used the Raw Material Data (RMD) database, which was discontinued in 2014⁴³. RMD was compiled by IntierraRMG⁴⁴, which was later acquired by SNL Metals & Mining (S&P Global). We therefore assume that the S&P dataset comprises the information contained in RMD and decide to use data from S&P Capital IQ Pro.

Our goal is to disaggregate q_i^r (ore production coefficients) by region of control. For each of the ten metals—copper, nickel, lead, zinc, tin, iron, aluminium, uranium, gold, and silver—we downloaded all the mines included in the S&P database. For each mine, we accessed its control share by owner and replaced the name of the owner with the nationality of the owner. For each commodity, this allowed to have access to the amount of production in each country by country of control. We then compared the national production levels with aggregates given by the British Geological Survey (BGS). When the sum of the S&P data for a given country and commodity was inferior to the BGS statistic, we added the difference to the S&P dataset and set the owner nationality as "unknown". The coverage rate of S&P compared with BGS is given in the sheet "S&P coverage" of the "coverage.xlsx" file in the FigShare associated with this article. This allowed to have a matrix K giving the metal production by region of control. K has rows that represent the country of control m in $\{1, \dots, 164\}$ and columns with the same dimensions as q_i . K is thus the matrix of elements K_i^{mr} and of dimension 164×19680 .

The disaggregation of metal production by region of control m (in percents) is therefore obtained by:

$$S_i^{mr} = K_i^{mr} \cdot \sum_m \widehat{K_i^{mr}}^{-1}$$

Where the $\widehat{}$ corresponds to a diagonalized vector.

We then use this matrix S_i^{mr} to disaggregate the ore production coefficients by region of control:

$$C_i^{mr} = S_i^{mr} \cdot \hat{q}_i^r$$

Using matrix C that contains information on both nationality of control and region of extraction, the consumption-based account of region z is thus:

$$CB^z = \sum_{ijk, mrs} C_i^{mr} L_{ij}^{rs} Y_{jk}^{sz}$$

Sankeys. In the Sankey diagram, we want to draw all flows related to country z, that is its consumption-based account plus its exports plus the amount of ores it controls via its mine ownership but that is neither embodied in its final consumption, nor extracted in z (i.e. not already considered in the exports of z). The data selected for the Sankey diagram of country z, such as Fig. 2b for Zambia, is thus given by:

$$D^z = CB^z + \underbrace{\sum_{ijk, ms(t \neq z)} C_i^{mz} L_{ij}^{zs} Y_{jk}^{st}}_{\text{exports of } z} + \underbrace{\sum_{ijk, (r \neq z)s(t \neq z)} C_i^{rz} L_{ij}^{rs} Y_{jk}^{st}}_{\text{other control of } z}$$

For the visualization of Fig. 2b, the first step is an aggregation by nationality of mine owners, so an aggregation over *m*. The second step is the production-based account by region; we aggregated the data over *r*. The third step is the consumption-based account by region; we aggregated the data over *t*. The fourth step is the consumption-based account by final demand category, we aggregated over *k*. The last step is the consumption-based account by final demand sector, we aggregated over *j*.

To aggregate regions, we used the concordance table available in SI 2.5 of ref. 21. The concordance table for final consumption sectors is available in the file “coverage.xlsx” in the FigShare associated with this article. The sectors building, civil engineering, electronic and electrical equipment, machinery and equipment, transport equipment, other manufactured goods, raw materials and “others” were selected as they were the main drivers of raw material use. An important part of these footprints corresponds to Gross Fixed Capital Formation (GFCF), which is in line with previous research showing that capital formation is driving raw material use^{45–48}.

The interactive application was then built using the open-access code from ref. 49. The first tab represents data without ownership, for users that want information on physical flows. The second one contains all the Sankeys with ownership data. A dropdown menu allows to select the country and another one to select the unit (either Mt or t/capita). A slider allows to select the year. It was not possible to make the Sankeys proportional from one year to another (e.g. make a Sankey of 2022 look 5 times bigger than one of 2005) due to an internal limit in plotly go Sankey, the python library used to build the Sankeys. The Sankeys in plotly have to fill a given amount of the figure (around 50%). This lower bound limits the relative size of two different Sankey diagrams to a factor of two.

Analysis. The ownership ratio (OR) is the share of a country’s mine production controlled domestically, divided by its total mine production. The ownership ratio of region z is:

$$OR^z = \frac{\sum_{ijk, st} C_i^{zz} L_{ij}^{zs} Y_{jk}^{st}}{CB^z}$$

This ownership ratio is compared with gross domestic product (GDP) per capita in Fig. 3. The GDP and population data were downloaded from the World Bank’s World Development Indicators database⁵⁰.

The aggregated dependency (AD) of region z on imports from a production perspective is $(1-\alpha)*100$, with α the total amount of ores produced by a country divided by the amount of ores embodied in its final

consumption:

$$AD_{prod}^z = \left(1 - \frac{\sum_{ijk, mst} C_i^{mz} L_{ij}^{zs} Y_{jk}^{st}}{CB^z}\right) \cdot 100$$

The “real” dependency (RD) considers only the part of the production of region z that ends up in the final consumption of region z:

$$RD_{prod}^z = \left(1 - \frac{\sum_{ijk, ms} C_i^{mz} L_{ij}^{zs} Y_{jk}^{sz}}{CB^z}\right) \cdot 100$$

The aggregated dependency (AD) of region z on imports from a control perspective is $(1-\alpha)*100$, with α the total amount of ores controlled by a country divided by the amount of ores embodied in its final consumption:

$$AD_{control}^z = \left(1 - \frac{\sum_{ijk, rst} C_i^{rz} L_{ij}^{rs} Y_{jk}^{st}}{CB^z}\right) \cdot 100$$

The “real” dependency (RD) considers only the part of the production of region z that ends up in the final consumption of region z:

$$RD_{control}^z = \left(1 - \frac{\sum_{ijk, rs} C_i^{rz} L_{ij}^{rs} Y_{jk}^{sz}}{CB^z}\right) \cdot 100$$

After calculating dependencies, it is usual to compile a risk indicator. However, most of the studies doing so are flawed and compiling it consistently with risk theory is a very complex task^{7,51}. Doing so is beyond the scope of the present analysis.

For the analysis of the impact control has on trade, the control of region B in region A is:

$$\text{Control}^{B \text{ in } A} = \sum_{ijk, st} C_i^{ba} L_{ij}^{as} Y_{jk}^{st}$$

The direct exports of ores from region A to region B is:

$$\text{Flow}_{direct}^{A \text{ to } B} = \sum_{ij, m} C_i^{ma} L_{ij}^{ab}$$

The exports of metals from region A to region B from ores extracted in region A is the amount of ores extracted in region A staying in region A during the first economic exchange, multiplied by the share of metal production (sector u) from A sent to B:

$$\text{Flow}_{metal}^{A \text{ to } B} = \left(\sum_{ij, m} C_i^{ma} T_{ij}^{aa} \right) \cdot \frac{\sum_j T_{uj}^{aa}}{\sum_{j, s} T_{uj}^{as}}$$

The indirect exports of metal ores from region A to region B is the part of region B’s consumption-based account that has been extracted in region A:

$$\text{Flow}_{indirect}^{A \text{ to } B} = \sum_{ijk, ms} C_i^{ma} L_{ij}^{as} Y_{jk}^{sb}$$

Data availability

The GLORIA MRIO Database is available for download on IELab’s website: <https://ielab.info/>. S&P Global IQ Pro is under payroll, but the aggregated country control file is available on the sheet named ‘table1_ownership_B_in_A’ of the in the file “table1_data.xlsx” in the following FigShare folder: <https://doi.org/10.6084/m9.figshare.28777013.v1>. All the other data used to draw the figures is also in that folder. The Sankeys are available on the web application at sankey-dashboard.inefficiency.org, and can be downloaded as a single compressed folder from the FigShare repository.

Code availability

The code associated with this publication is available on GitHub: <https://github.com/baptiste-an/Mapping-metal-flows-sankeys>.

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B.A.: Conceptualization, Methodology, Formal Analysis, Writing—Original Draft K.C.B.: Methodology, Validation, Writing—Review & Editing M.H.: Methodology, Validation, Writing—Review & Editing A.K.: Methodology, Validation, Writing—Review & Editing J.C.: Supervision, Project Administration, Funding Acquisition, Methodology, Validation, Writing—Review & Editing.

Competing interests

The authors declare no competing interests.

Additional information

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